## 4.4 THE THIN LENS LO 2

A medium bounded by two spherical refracting surfaces is referred to as a spherical lens. If the thickness of such a lens (shown as $t$ in Fig. 4.5) is very small compared to object and image distances, and to the radii of curvature of the refracting surfaces then the lens is referred to as a thin spherical lens. In general, a lens may have non-spherical refracting surfaces (e.g., it may have cylindrical surfaces). However, most lenses employed in optical systems have spherical refracting surfaces. Therefore, we will simply use the term 'lens' to imply a spherical lens. Different types of lenses are shown in Fig. 4.6. The line joining the centers of curvature of the spherical refracting surfaces is referred to as the *axis* of the lens.

In this section, we will consider the paraxial image formation by a thin lens. The corresponding considerations for a thick lens will be discussed in Problem 4.6.

We consider a point object $O$ on the axis of a (thin) lens as shown in Fig. 4.5. The lens is placed in a medium of refractive index $n_1$ and the refractive index of the material of the lens is $n_2$. Let $R_1$ and $R_2$ be the radii of curvature of the left and right surfaces of the lens; for the lens shown in Fig. 4.5, $R_1$ is positive and $R_2$ is negative. In order to determine the position of the image, we will consider successive refractions at the two surfaces; the image formed by the first surface is considered as the object (which may be real or virtual) for the second surface. Thus, if the second refracting surface had not been there, the image of the point $O$ would have been formed at $Q$ whose position (given by $v'$) is determined from Eq. (4.8) (see also Eq. (4.5))

$$\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \qquad \textbf{(4.8)}$$

where $u$ is the object distance which is negative for the object point $O$ shown in the figure. Obviously if $v'$ is positive, then the point $Q$ lies to the right of the surface and if $v'$ is negative then $Q$ lies to the left of the surface. The point $Q$ now acts as the (virtual) object for the second refracting surface and the final image is formed at $I$ whose position is determined from the equation

$$\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2} \qquad \textbf{(4.9)}$$

In Eqs. (4.8) and (4.9) the distances are measured from the center of the lens $P$; this is justified because the lens has been assumed to be thin. Adding Eqs. (4.8) and (4.9), we get

$$\frac{1}{v} - \frac{1}{u} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad \textbf{(4.10)}$$

where

$$n \equiv \frac{n_2}{n_1}$$

Equation (4.10) is known as the *thin lens formula* and is usually written in the form

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \qquad \textbf{(4.11)}$$

where $f$, known as the focal length of the lens, is given by

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad \textbf{(4.12)}$$

For a lens placed in air (which is usually the case), $n > 1$ and if $[(1/R_1) - (1/R_2)]$ is a positive quantity then the focal length is positive and the lens acts as a converging lens [see Fig. 4.7(a)]. Similarly, if $(1/R_1) - (1/R_2)$ is a negative quantity then the lens acts as a diverging lens (see Fig. 4.7(b)). However, if the double convex lens is placed in a medium whose refractive index is greater than that of the material of the lens, then the focal length becomes negative and the lens acts as a diverging lens [see Fig. 4.7(c)]; similarly for the double concave lens [see Fig. 4.7(d)].

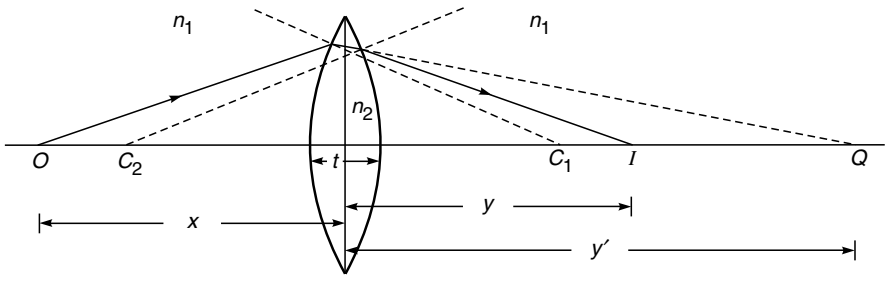



**Fig. 4.5** Image formation by a thin lens. The line joining the two centers of curvature is known as the axis of the lens [$u = -x$, $v' = y'$ $v = y$].

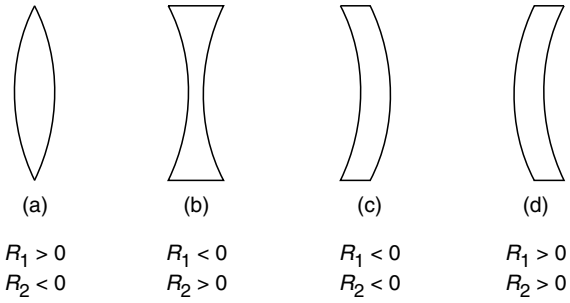



**Fig. 4.6** Signs of $R_1$ and $R_2$ for different lens types.

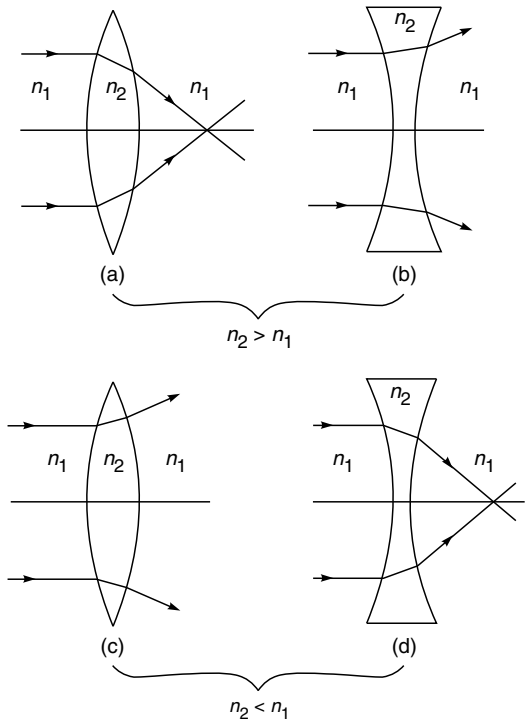



**Fig. 4.7** (a) and (b) correspond to the situation when the refractive index of the material of the lens is greater than that of the surroundings and therefore a biconvex lens acts as a converging lens and a biconcave lens acts as a diverging lens. (c) and (d) correspond to the situation when the refractive index of the material of the lens is smaller than that of the surrounding medium and therefore a biconvex lens acts as a diverging lens and a biconcave lens as a converging lens.

## 4.5 THE PRINCIPAL FOCI AND FOCAL LENGTHS OF A LENS

For a converging lens, the *first principal focus* is defined as the point (on the axis) such that a ray passing through that point will, after refraction through the lens, emerge parallel to the axis—see ray 1 in Fig. 4.8(a); the point $F_1$ is the first principal focus. For a diverging lens, the ray which (in the absence of the lens) would have passed through the first principal focus emerges, after refraction by the lens, as a ray parallel to the axis—see ray 1 in Fig. 4.8(b). The point $F_1$ is the **first principal focus** and its distance from the lens (denoted by $f_1$) is known as the **first focal length** of the lens. Obviously, $f_1$ is negative for a converging lens and positive for a diverging lens.

We next consider a ray which travels parallel to the axis [see ray 2 in Figs. 4.8(a) and (b)]. For a converging lens the point at which the ray will intersect the axis [shown as $F_2$ in Fig. 4.8(a)] is known as the **second principal focus** of the lens. Similarly, for a diverging lens, the point at which the ray would have intersected the axis (if produced backwards) is the second principal focus [see the point $F_2$ in Fig. 4.8(b)]. The distance of the second principal focus from the lens is known as the **second focal length** and is denoted by $f_2$. As can be seen from Fig. 4.8, $f_2$ is positive for a converging lens and negative for a diverging lens.

For a thin lens placed in a medium such that the refractive indices on both sides of the lens are the same ($n_3 = n_1$ in Fig. 4.8), the values of $f_1$ and $f_2$ can be readily obtained by considering the thin lens formula [see Eq. (4.10)] and one gets

$$\frac{1}{f_2} = -\frac{1}{f_1} = \left[(n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)\right] = \frac{1}{f} \qquad \textbf{(4.13)}$$

However, if $n_3 \neq n_1$ then the thin lens formula assumes the form as given in Eq. (4.14) (see also Problem 4.2):

$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2} \qquad \textbf{(4.14)}$$

Now, when $v = \infty$, $u = f_1$ (ray 1 in Fig. 4.8), we have

$$\frac{1}{f_1} = -\frac{1}{n_1}\left[\frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}\right] \qquad \textbf{(4.15)}$$

Similarly, when $u = -\infty$, $v = f_2$ (ray 2 in Fig. 4.8) we have

$$\frac{1}{f_2} = \frac{1}{n_3}\left[\frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2}\right] \qquad \textbf{(4.16)}$$

Once we know $f_1$ and $f_2$ (and therefore the positions of the first and second principal foci) the (paraxial) image can be graphically constructed from the following rules: